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COMET FWC 071

Implementation of a Boolean Expression Using Arduino UNO, SN74LS47N Decoder and Seven Segment Display

To implement the Boolean expression obtained from the given logic circuit using Arduino UNO, SN74LS47N BCD-to-Seven Segment Decoder and a Common Anode Seven Segment Display.

Apparatus Required

- Arduino UNO
- SN74LS47N BCD-to-Seven Segment Decoder
- Common Anode Seven Segment Display
- Breadboard
- Connecting Wires
- 220Ω Resistors
- USB Cable

Logic Diagram

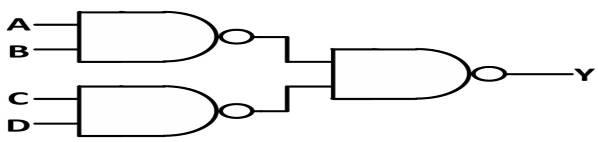

Figure 1: Given Logic Circuit

$$Y = AB + CD \tag{1}$$

The Boolean expression is evaluated by the Arduino UNO. The resulting output is sent as BCD data to the SN74LS47N decoder, which drives a Common Anode Seven Segment Display.

Truth Table

Table 1: Truth Table of the Boolean Expression

A	B	C	D	Y = AB + CD
0	0	0	0	0
1	1	0	0	1
0	1	0	1	0
0	0	1	1	1

Pin Connections

Arduino UNO to SN74LS47N

Table 2: Arduino to Decoder Connections

Arduino Pin	SN74LS47N Pin heightD9
A (Pin 7)	
D10	B (Pin 1)
D11	C (Pin 2)
D12	D (Pin 6)
5V	VCC (Pin 16)
GND	GND (Pin 8)

SN74LS47N Control Pins

Table 3: Control Pin Connections

SN74LS47N Pin	Connection
LT (Pin 3)	+5V
RBI (Pin 4)	+5V
BI/RBO (Pin 5)	+5V

SN74LS47N to Seven Segment Display

Table 4: Decoder to Display Connections

SN74LS47N Output	Seven Segment Segment
a (Pin 13)	a
b (Pin 12)	b
c (Pin 11)	c
d (Pin 10)	d
e (Pin 9)	e
f (Pin 15)	f
g (Pin 14)	g

Common Anode Connections

Table 5: Common Anode Display Connections

Display Pin	Connection
Common Anode Pin 1	+5V
Common Anode Pin 2	+5V

Observations

Table 6: Experimental Verification

A	B	C	D	Displayed Output
0	0	0	0	0
1	1	0	0	1
0	1	0	1	0
0	0	1	1	1

The output sequence observed on the seven segment display is

$$0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \tag{2}$$

Result

The Boolean expression

$$Y = AB + CD \tag{3}$$

was successfully implemented using Arduino UNO, SN74LS47N BCD-to-Seven Segment Decoder and a Common Anode Seven Segment Display. The experimental results matched the expected truth table outputs.